Practical No. 05

a) Develop an Android Application using ListView with ArrayAdapter and implement item click handling.

Project Name: ListViewDemo

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="409dp"
        android:layout_height="729dp"
        android:orientation="vertical"
        tools:layout_editor_absoluteX="1dp"
        tools:layout_editor_absoluteY="1dp">

        <TextView
            android:id="@+id/textView"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:paddingBottom="10dp"
            android:text="Programming Language"
            android:textSize="24sp"
            android:textStyle="bold" />

        <ListView
            android:id="@+id/listView"
            android:layout_width="match_parent"
            android:layout_height="match_parent" />

    </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

Filename: MainActivity.java

```
package com.example.listviewdemo;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    ListView listView;

    String[] languages = {
            "Java",
            "Python",
            "C",
            "C++",
            "Kotlin",
            "PHP",
            "Javascript",
            "Swift"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);

        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                languages
        );

        listView.setAdapter(adapter);

        listView.setOnItemClickListener(
                new AdapterView.OnItemClickListener() {

                    @Override
                    public void onItemClick(
                            AdapterView<?> parent,
                            View view,
                            int position,
                            long id
                    ) {
                        String item = languages[position];

                        Toast.makeText(
                                MainActivity.this,
                                "Selected: " + item,
                                Toast.LENGTH_SHORT
                        ).show();
                    }
                }
        );
```

```
        ViewCompat.setOnApplyWindowInsetsListener(
                findViewById(R.id.main),
                (v, insets) -> {

                    Insets systemBars = insets.getInsets(
                            WindowInsetsCompat.Type.systemBars()
                    );

                    v.setPadding(
                            systemBars.left,
                            systemBars.top,
                            systemBars.right,
                            systemBars.bottom
                    );

                    return insets;
                }
        );
    }
}
```

b) Develop an Android application using GridView with Custom Adapter and image selection handling.

Project Name: GridViewDemo

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="409dp"
        android:layout_height="729dp"
        android:orientation="vertical"
        tools:layout_editor_absoluteX="1dp"
        tools:layout_editor_absoluteY="1dp">

        <TextView
            android:id="@+id/textView2"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:paddingBottom="10dp"
            android:text="Image Gallery"
            android:textSize="24sp"
            android:textStyle="bold" />

        <GridView
            android:id="@+id/gridView"
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:numColumns="2"
            android:verticalSpacing="10dp"
            android:horizontalSpacing="10dp"
            android:stretchMode="columnWidth" />

    </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

Filename: grid_item.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="10dp">

        <ImageView
            android:id="@+id/imageView"
            android:layout_width="120dp"
            android:layout_height="120dp"
            android:scaleType="centerCrop" />

        <TextView
            android:id="@+id/textView"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:textSize="18sp"
            android:paddingTop="5dp" />

    </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

Filename: CustomAdapter.java

```
package com.example.grideviewdemo;

import android.content.Context;

import android.view.LayoutInflater;

import android.view.View;

import android.view.ViewGroup;

import android.widget.BaseAdapter;

import android.widget.ImageView;

import android.widget.TextView;



public class CustomAdapter extends BaseAdapter {



    Context context;

    String[] imageNames;

    int[] images;

    LayoutInflater inflater;
```

```
public CustomAdapter(Context context, String[] imageNames, int[] images) {
    this.context = context;
    this.imageNames = imageNames;
    this.images = images;

    inflater = LayoutInflater.from(context);
}

@Override
public int getCount() {
    return imageNames.length;
}

@Override
public Object getItem(int position) {
    return imageNames[position];
}

@Override
public long getItemId(int position) {
    return position;
}

@Override
public View getView(int position, View convertView, ViewGroup parent) {

    View view = inflater.inflate(
            R.layout.grid_item,
            parent,
            false
    );
```

```
        ImageView imageView = view.findViewById(R.id.imageView);

        TextView textView = view.findViewById(R.id.textView);



        imageView.setImageResource(images[position]);

        textView.setText(imageNames[position]);



        return view;

    }

}
```

Filename: MainActivity.java

```
package com.example.grideviewdemo;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.GridView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    GridView gridView;

    String[] imageNames = {
            "Human",
            "Food",
            "Tree",
            "Animal"
    };

    int[] images = {
            R.drawable.image1,
            R.drawable.image2,
            R.drawable.image3,
            R.drawable.image4
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        gridView = findViewById(R.id.gridView);

        CustomAdapter adapter = new CustomAdapter(
                this,
                imageNames,
                images
        );

        gridView.setAdapter(adapter);
```

```
        gridView.setOnItemClickListener(
                new AdapterView.OnItemClickListener() {

                    @Override
                    public void onItemClick(
                            AdapterView<?> parent,
                            View view,
                            int position,
                            long id
                    ) {
                        Toast.makeText(
                                MainActivity.this,
                                "Selected: " + imageNames[position],
                                Toast.LENGTH_SHORT
                        ).show();
                    }
                }
        );

        ViewCompat.setOnApplyWindowInsetsListener(
                findViewById(R.id.main),
                (v, insets) -> {

                    Insets systemBars = insets.getInsets(
                            WindowInsetsCompat.Type.systemBars()
                    );

                    v.setPadding(
                            systemBars.left,
                            systemBars.top,
                            systemBars.right,
                            systemBars.bottom
                    );

                    return insets;
                }
        );
    }
}
```

c) Develop an Android application implementing RecyclerView as a modern alternative to ListView

Project Name: RecyclerViewDemo

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginVertical="50dp"
        android:text="RecyclerViewDemo"
        android:textAlignment="center"
        android:textSize="20dp"
        android:paddingBottom="10dp"
        android:textStyle="bold" />

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recyclerView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

Filename: item_layout.xml

```
<?xml version="1.0" encoding="utf-8"?>

<TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/txtItem"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="20sp"
    android:padding="15dp" />
```

Filename: MyAdapter.java

```
package com.example.recyclerviewdemo;

import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;
```

```
public class MyAdapter extends RecyclerView.Adapter<MyAdapter.ViewHolder> {

    String[] data;

    public MyAdapter(String[] data) {
        this.data = data;
    }

    public static class ViewHolder extends RecyclerView.ViewHolder {

        TextView txtItem;

        public ViewHolder(@NonNull View itemView) {
            super(itemView);

            txtItem = itemView.findViewById(R.id.txtItem);
        }
    }

    @NonNull
    @Override
    public ViewHolder onCreateViewHolder(
            @NonNull ViewGroup parent,
            int viewType
    ) {

        View view = LayoutInflater.from(parent.getContext())
                .inflate(
                        R.layout.item_layout,
                        parent,
                        false
                );

        return new ViewHolder(view);
    }

    @Override
    public void onBindViewHolder(
            @NonNull ViewHolder holder,
            int position
    ) {

        holder.txtItem.setText(data[position]);

        holder.itemView.setOnClickListener(v -> {

            Toast.makeText(
                    v.getContext(),
                    "Selected: " + data[position],
                    Toast.LENGTH_SHORT
            ).show();

        });
    }

    @Override
    public int getItemCount() {
        return data.length;
    }
}
```

Filename: MainAcitivty.java

```
package com.example.recyclerviewdemo;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

public class MainActivity extends AppCompatActivity {

    RecyclerView recyclerView;

    String[] languages = {
            "Java",
            "Python",
            "C",
            "C++",
            "koylin",
            "PHP",
            "JavaScript",
            "Swift",
            "Go",
            "Ruby"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        recyclerView = findViewById(R.id.recyclerView);

        recyclerView.setLayoutManager(
                new LinearLayoutManager(this)
        );

        MyAdapter adapter = new MyAdapter(languages);

        recyclerView.setAdapter(adapter);
    }
}
```

Filename: build.gradle (:app)

```
plugins {
    alias(libs.plugins.android.application)
}

android {
    namespace 'com.example.recyclerviewdemo'
    compileSdk 36

    defaultConfig {
        applicationId "com.example.recyclerviewdemo"
        minSdk 30
        targetSdk 36
        versionCode 1
        versionName "1.0"

        testInstrumentationRunner "androidx.test.runner.AndroidJUnitRunner"
```

```
    }

    buildTypes {
        release {
            minifyEnabled false

            proguardFiles getDefaultProguardFile(
                    'proguard-android-optimize.txt'
            ),
            'proguard-rules.pro'
        }
    }

    compileOptions {
        sourceCompatibility JavaVersion.VERSION_11
        targetCompatibility JavaVersion.VERSION_11
    }
}

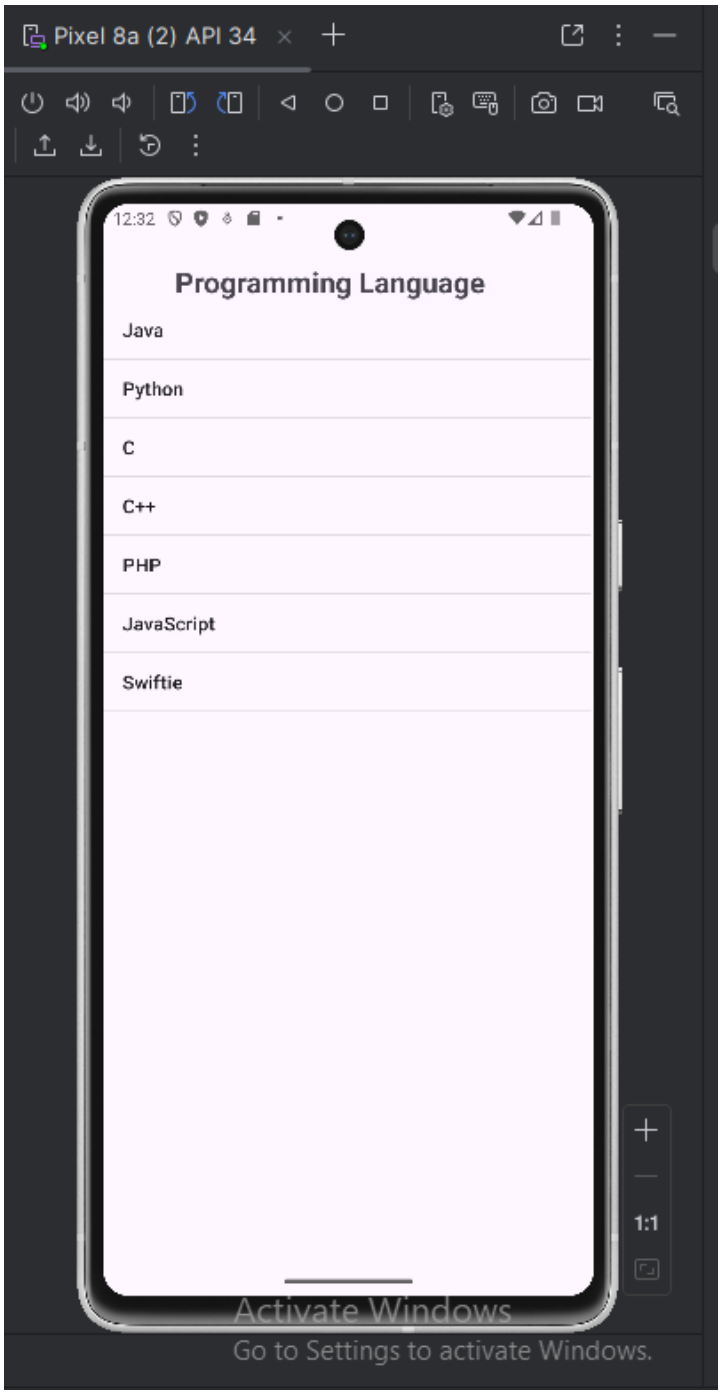
dependencies {
    implementation libs.appcompat
    implementation libs.material
    implementation libs.activity
    implementation libs.constraintlayout

    testImplementation libs.junit

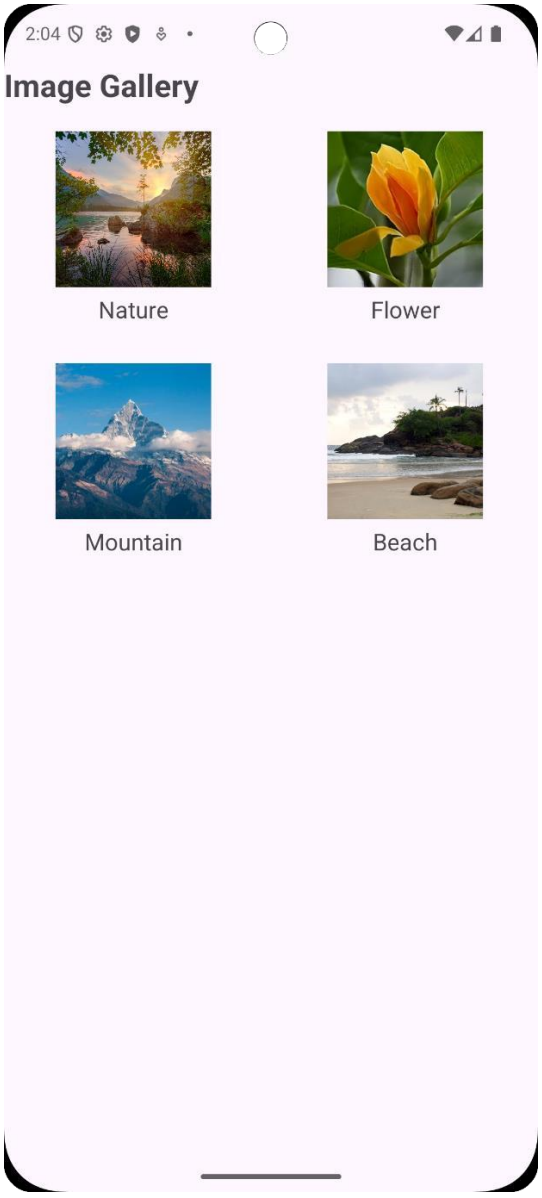
    androidTestImplementation libs.ext.junit
    androidTestImplementation libs.espresso.core

    implementation("androidx.recyclerview:recyclerview:1.3.2")
}
```

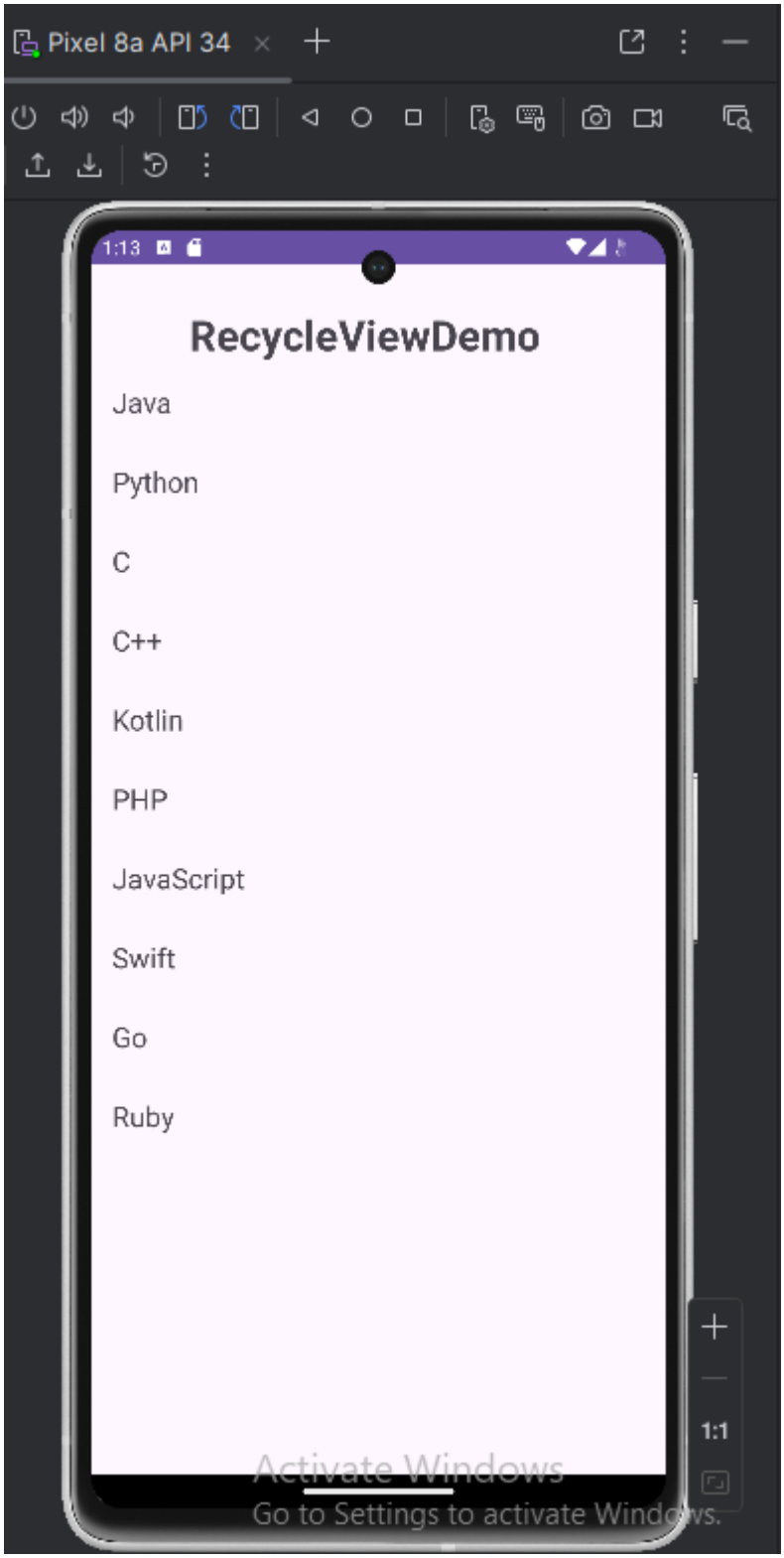
a) Output:


Pixel 8a (2) API 34
12:32
Programming Language
Java
Python
C
C++
PHP
JavaScript
Swiftie
1:1
Activate Windows
Go to Settings to activate Windows.

b) Output:



c) Output:


Pixel 8a API 34
1:13
RecycleViewDemo
Java
Python
C
C++
Kotlin
PHP
JavaScript
Swift
Go
Ruby
1:1
Activate Windows
Go to Settings to activate Windows.